

Building a LaTeX Report Generation Skill

A context-aware document generator for Claude

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Abstract

This report summarizes the design of a Claude skill for generating professional documents as LaTeX source, compiled directly to PDF. The skill infers document type and structure from conversational context rather than requiring the user to specify formatting up front, and draws its content from the actual work discussed in the session.

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1 Introduction

The goal was a single, general-purpose skill capable of producing any kind of professional LaTeX document – project reports, technical papers, and memos – without the user needing to specify layout, packages, or structure each time.

2 Approach

2.1 Template Strategy

Rather than generating LaTeX from a blank page every time, the skill bundles three starter templates covering the most common document shapes, each pre-loaded with sensible packages and a placeholder structure:

- `project-report.tex` – title page, abstract, TOC, numbered sections
- `tech-paper.tex` – IEEE-style two-column paper with abstract and references
- `memo.tex` – short header-block document, no TOC

2.2 Verification

Each template was compiled with `pdflatex` to confirm it builds cleanly before being bundled:

```
1 $ pdflatex -interaction=nonstopmode project-report.tex
2 $ pdflatex -interaction=nonstopmode tech-paper.tex
3 $ pdflatex -interaction=nonstopmode memo.tex
```

Listing 1: Compile check

One issue was found and fixed during testing: an underscore in a placeholder header string was being parsed as a math-mode subscript, which broke compilation. It was replaced with a space.

3 Results

Table 1: Templates bundled with the skill

Template	Structure	Typical use
<code>project-report.tex</code>	Title page + TOC	Status/KT/engineering reports
<code>tech-paper.tex</code>	Two-column, IEEE	Research/conference papers
<code>memo.tex</code>	Header block only	Short proposals, memos

All three templates compile without errors on a stock TeX Live install, with one caveat: `IEEEtran.cls` is not present by default and must be installed via `apt-get install -y texlive-publishers` on a fresh sandbox.

4 Next Steps

Package the skill and hand it to the user for review; iterate on template structure based on real documents generated with it.

5 Conclusion

The skill is ready for a first real-world test run.